



Rakesh Kumar Reddy Pothireddy

M.Tech Computer Science & Engineering / NIS - TA

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Degree	University/Institute	Year	CGPA/Marks(%)
M.Tech Computer Science & Engineering / NIS - TA	IIT Hyderabad	2026	8.32
B.Tech Electronics & Communication Engineering	IIIT Sricity	2023	8.52
XII (Telangana State Board)	Deeksha Junior College Hyderabad	2019	98.60%
X (CBSE)	JNV Kadapa	2017	10.00

PROJECTS

Generative AI for Infographic Synthesis

MTech Capstone Project — Ongoing

- Explored **Stable Diffusion XL + ControlNet** for controllable infographic generation; implemented experiments and evaluated limitations for structured, multi-layer content.
- Researched **BizGen (2024)** and aligned project direction with **CLIP-based encodings, layout-aware prompting, and dense layer decomposition** for article-level infographic synthesis.
- Currently designing a pipeline for **layered, editable outputs (SVG/PSD)** by integrating diffusion models with segmentation (SAM/CLIP) and raster-to-vector conversion tools.

Android Malware Detection (Static + ML Pipeline)

Aug–Sep 2025

- Designed and implemented a **static analysis and machine learning pipeline** for Android malware detection, processing ~60K benign and malware APKs.
- Extracted and analyzed key features (permissions, manifest components, DEX-level API calls), applied **EDA and chi-square feature selection** to reduce dimensionality.
- Trained and evaluated ML models (**Logistic Regression, Random Forest, XGBoost**), achieving **>98% accuracy**; built an inference pipeline for classifying unseen APKs.

Tech stack: Python, Androguard, Scikit-learn, XGBoost, Pandas

Ping Suite for Network Performance Analysis

Sep-2024

- Built Python-based **UDP, TCP, and ICMP ping tools** to measure RTT, packet loss, and host availability.
- Developed a **multithreaded TCP server** with timeout and exception handling for concurrent client pings.
- Simulated real-world network degradation using **Linux tc netem** to test performance under 20% packet loss.

RELEVANT COURSES

- Foundations of Machine Learning
- Deep Learning
- Advanced Computer Networks
- Practical Malware Analysis

SKILLS

Technical Skills : Machine Learning, Deep Learning, Socket Programming, MySQL, Data structures, OOPs

Programming Languages : C, C++, Python.

POSITIONS OF RESPONSIBILITY

- Teaching Assistant for the course Introduction to Programming(ID1063) Aug 2024 - Nov 2024
- Teaching Assistant for the course Algorithms(CS2443) Jan 2025 - Apr 2025
- Teaching Assistant for the course Data Structures and Applications(ID2230) Aug 2025 - present

SCHOLASTIC ACHIEVEMENTS

- Achieved the **highest CGPA of the batch** in the **first semester**, demonstrating academic excellence.
- Secured an All India Rank (AIR) of 370 in GATE CS 2024.
- Achieved LeetCode Contest Rating 1732 (Top 10.8% globally)
- Secured an All India Rank (AIR) of 871 in GATE DA 2024.
- Secured 97.43 percentile in JEE Main 2019.

EXTRACURRICULAR

- Won **Bronze** in **Badminton** representing Kalam Hostel at Milan, the annual sports event of IITH.
- PG Mentor of *Sunshine Wellbeing Club*, guiding peers and organizing initiatives to promote mental health and wellbeing.